Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years using **C++**, **Python**, Java, and JavaScript across automotive simulation, PC/web applications, data acquisition, and automated testing. Experienced in requirements analysis, debugging, technical documentation, reporting, and reliable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed a **Python** and web-based application for 500+ users with data downloads, operational dashboards, access controls, and monitored execution, reducing response latency to 350 milliseconds through systematic performance analysis.
- Documented and tested asynchronous software processing 1,000+ jobs through defined states, structured logs, fault isolation, and recovery procedures, increasing throughput threefold while preventing cascading failures under production load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built PC-accessible web interfaces and Python services handling 10,000+ daily requests, implementing secure data ingestion, validation, diagnostic status views, and repeatable functional tests while maintaining stable peak-load behavior.
- Developed a **multithreaded** data-processing pipeline for 1,000+ files using ThreadPoolExecutor, progress reporting, status visibility, fault recovery, and repeatable defect reproduction, reducing end-to-end processing turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **automotive simulation and verification pipeline** processing 100,000+ vehicle trajectory records, improving planning quality 18% through reproducible test scenarios, automated validation, performance tracing, and controlled configuration.
- Evaluated vehicle and sensor behavior across closed-loop scenarios, improving simulation stability 15% while documenting expected outputs, isolating failures, comparing diagnostic results, and debugging interactions between software components.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered PC-based cross-platform software with automated executable builds and VM testing across Windows, Linux, and macOS, maintaining 20+ test suites, coverage reports, CSV results, and detailed failure logs.
- Analyzed requirements and enhanced object-oriented integration modules using Python, Java, SQL, and REST APIs, improving response times 30% and processing efficiency 68% through regression testing and tracked corrections.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed web-based TypeScript and Node.js software across eight service modules, implementing defined interfaces, data-conversion workflows, reports, and real-time status updates spanning more than two million operational records.
- Automated functional, integration, and regression testing with Jenkins, Docker, Jest, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving software release verification and issue traceability.

Skills

Languages & Applications: C++, Python, Java, JavaScript, TypeScript, C++ object-oriented programming, PC applications, web applications, REST APIs, multithreading
Automotive & Data: Autonomous-driving simulation, vehicle trajectories, sensor data, data acquisition, data conversion, operational dashboards, status reporting, diagnostics interest
Testing & Debugging: PyTest, Mockito, Jest, Selenium, functional testing, integration testing, regression testing, automated tests, defect reproduction, fault isolation, logging, monitoring
Development & Documentation: Git, Docker, Jenkins, Maven, CI/CD, IDEs, source control, issue tracking, requirements documentation, test results, technical reports, user-facing guidance

Projects

CPU-Scheduling Simulator | Live | *C++, React, Operating Systems, EJ-2*
- Implemented six deterministic CPU-scheduling algorithms in **C++** with React controls and Gantt visualization, validating 1,000+ process combinations while reducing average turnaround time 25% and debugging time 20%.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

FinNexus | Live | *Real-Time Systems, WebSockets, Kubernetes, MongoDB*
- Built a real-time platform supporting 50,000+ users and one million transactions, reliably integrating WebSocket updates, failure monitoring, Kubernetes deployment, automated releases, and performance-tuned services under high-volume workloads.